Sankranti Joshi
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Nationality: British | OCI Holder (India) | IEC work permit (Canada)

Professional Summary

- Machine Learning Engineer with 6+ years of experience in deep learning, computer vision, NLP, and MLOps.
- Strong background in AI research, model deployment, and high-performance computing.
- Expertise in Transformers, CNNs, contrastive learning, and generative models. Proven track record of building scalable ML solutions in fintech, healthcare, and geospatial domains.

Technical Skills

- **Programming & Tools**: Python, PyTorch, TensorFlow, Bash, Docker, PyTorch-Lightning, Weights & Biases, Slurm, HPC, Poetry, scikit-learn, XGBoost, Hugging Face, DeepSpeed, Kubernetes (basic)
- **Machine Learning & AI**: Transformers, CNNs, RNN/LSTMs, MAE, VAE, Language Models, Contrastive Learning, Diffusion Models
- **Cloud & MLOps**: AWS (SageMaker, EC2, RDS), Azure, CI/CD, Model Versioning, Feature Engineering

Industry & Research Experience

Senior Machine Learning Engineer | (July 2025 – Present) | Playstation

- Working on AutoQA problems and models.

Senior Machine Learning Engineer | (March 2025 – May 2025) | Self-employed contract

- Worked for lean health startup (Digehealth) as the sole ML engineer
- Built ML training and inference pipelines from scratch under limited infrastructure and noisy data conditions.
- Boosted YOLO-based audio-segmentation performance by 15+ mAP points through data processing, noise cancellation and hyperparameter tuning.
- Implemented reproducible training workflows on Azure ML with scalable, serverless compute.
- Acted as the sole ML and infra owner in a founder-led startup, while engaging directly with Microsoft engineers to resolve GPU driver issues.
- Factor analysis between YOLO predictions and downstream gut health related tasks.
- Contract concluding 30th May 2025 following a mid-project change in funding commitments

PhD Researcher (Earth Observation & ML) | The University of Edinburgh, UK (Oct 2023 – Mar 2025)

- Developed multimodal contextual dataset combining Landsat and Sentinel data, migrated to Amazon Open Data Sponsorship.
- Developed distributed training pipelines and docker files to train on HPC and Kubernetes clusters.
- Designed self-supervised MAE-based regression models for domain generalization in Earth Observation.
- Investigated large-scale foundation models for geospatial vision tasks.
- Withdrew due to unworkable supervisory situation.

Applied Science Intern (Internship as a requirement of the CDT) | Roku, Cambridge, UK (Jul 2024 – Sep 2024)

- Generated synthetic datasets using Stable Diffusion, ControlNet, inpainting/outpainting to enhance YOLOX model training.

- Developed RAG pipelines that generate prompts for generating an finetuning prompts for diffusion models.
- Led test/train dataset distribution strategy, improving model generalization.

Senior AIML Platform Engineer | GSK, London, UK (May 2023 - Oct 2023)
- Designed scalable ML pipelines for bio-model deployment on Azure HPC & cloud services.
- Developed an end-to-end DNA sequence classification pipeline, showcasing ML platform capabilities, and investigated fine-tuning, pruning and compression techniques.
- Transitioned to PhD program after internal role shift toward platform engineering.

Senior Machine Learning Engineer | Informa Markets, London, UK (Sep 2022 – Mar 2023)
- Developed LDA-guided BERT (LLM models) for document embeddings, improving NLP model accuracy by 25%.
- Built MLOps infrastructure (data versioning, model versioning, dependency management, CI/CD).
- Engineered dynamic pricing forecasting models using advanced data pipelines.

Senior Associate I, Data Science | AIG, London, UK (June 2019 – Aug 2022)
- Promoted from Machine Learning Engineer to Senior Associate I, Data Science (Apr 2021).
- Developed and maintained production repository for training and deployment of time-series forecasting model.
- Built geospatial Django pipelines, migrating legacy systems to DVC-based versioning.
- Developed end-to-end feature-engineering pipelines in Python with MLflow integration, enabling fully reproducible data transformations and experiment tracking.
- Refactored existing MXNet implementation of DeepAR-based time-series forecasting for tailored use-case.
- Left following department divestiture; Declined transition to acquiring company.

Graduate Software Developer (NLP Research) | Sage Group Plc, Newcastle, UK (Feb 2018 – Mar 2019)
- Developed a linguist-verified ground-truth generation pipeline for user chat data, ensuring high-quality annotations.
- Engineered and benchmarked custom NLP models (unsupervised methods, memory-augmented LSTMs, CRFs) against RASA and Alterra, driving measurable improvements in FAQ bot accuracy.
- Moved on to pursue production-focused ML work and growth opportunities.

<u>Key Projects & Contributions</u>
- **Open Data Contribution**: Created an open-source, multi-sensor satellite dataset, now part of AWS Open Data Registry.
- **Deep Learning for Vision**: Implemented MAE-based contrastive and self-supervised learning models for domain adaptation in geospatial imagery.
- **Synthetic Data Generation**: Applied Diffusion models to generate high-quality synthetic datasets for AI training.
- **MLOps with CI/CD and reproducibility**: Built automated ML pipelines with CI/CD and reproduction

<u>Education</u>
- **Master of Science in AI,** The University of Edinburgh, UK (2016 - 2017)
- **Master of Technology in Software Engineering**, Manipal Institute of Technology, India (2012 - 2014)
- **Bachelor of Technology in Computer Engineering**, Mody Institute of Technology and Science, India (2008 - 2012)